Modelling Upstream Plasma Formation in Heaterless Hollow Cathodes

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Abstract: This study investigates the effect of inlet diameter on upstream plasma formation in heaterless hollow cathodes (HHC), using a 2D axisymmetric fluid plasma model implemented in COMSOL Multiphysics. The model HHC has an open-end emitter orificed keeper arrangement. Two inlet diameters are simulated. Results show that a narrower inlet, approaching the diameter of the keeper orifice, effectively eliminates plasma formation in the gas inlet. The main plasma suppression mechanism identified is increased ion losses to the inlet walls.

Nomenclature

CMS	=	COMSOL Multiphysics software
D_{in}	=	inlet diameter, m
D_{ko}	=	keeper orifice diameter, m
E	=	electric field, V/m
e	=	unit charge, 1.602×10^{-19} C
I_a	=	anode current, A
I_k	=	keeper current, A
$\vec{J}_e$	=	electron current density, A/m^2
$j_{e,em}$	=	thermionic current density, A/m^2
n_e	=	electron density, m^{-3}
p	=	pressure, Pa
p_e	=	electron pressure, Pa
SEE	=	secondary electron emission coefficient
T_e	=	electron temperature, eV
T_{em}	=	emitter insert temperature, K
$V_{k,exp}$	=	keeper potential measured in experiment, V
$V_{k,sim}$	=	keeper potential calculated in simulation, V
γ_i	=	ion induced secondary electron emission coefficient
Γ_t	=	thermal electron emission flux, m^{-2}s^{-1}
$\Delta\varepsilon$	=	energy loss from reaction, eV

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I.Introduction

Heaterless hollow cathodes (HHC) are types of thermionic hollow cathodes that are designed to utilize gas breakdown followed by a glow discharge in order to heat an emitter insert to thermionic emission temperatures.[1] An HHC necessitates a carefully designed emitter-keeper cavity; typically, two such arrangements are implemented: 1) Orificed emitter which is the common configuration; 2) the open-end emitter orificed-keeper (OEEOK) configuration which is unique to HHC designs. In HHC, the control of internal pressure distribution is understood to be critical to the ignition and heating processes, as well as affecting possible transitions of the discharge into cathode arcing. Arcing in turn can cause significant damage to the cathode electrodes and limit both the number of ignitions and the cathode life. Some authors[2] recommend the use of a narrow inlet, relative to the diameter of the emitter insert cavity, to eliminate long-path-length Paschen breakdown from the keeper to the gas inlet.

In this paper we explore the influence of inlet diameter on upstream plasma formation in HHC by performing fluid plasma modeling of such a cathode.

II. Approach

To investigate the spatial characteristics of plasma parameters in a HHC, under different inlet diameters, this study employs a two-dimensional axisymmetric fluid model. The HHC configuration modeled is OEEOK, as shown in Fig. 1.

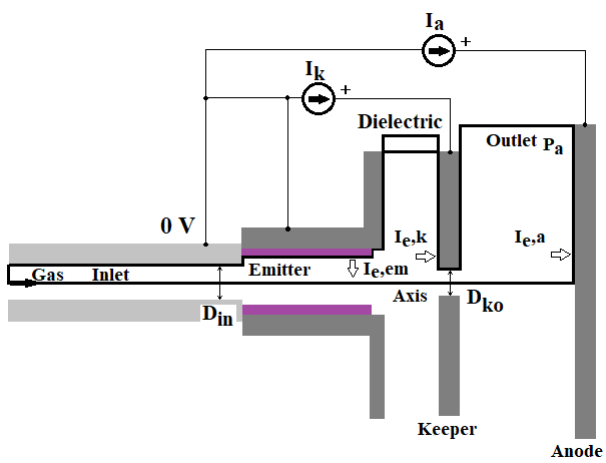

Figure 1. Schematic of OEEOK HHC model showing the computational region (upper half - surrounded by a black line bounded by the axis of symmetry).

The model is implemented using a COMSOL Multiphysics software (CMS).[3] The CMS was previously validated against experimental data and other hollow cathode models.[4,5] Primary inputs to the model include anode and keeper currents, emission thermionic flux, and gas mass flow rate. Additional inputs include plasma transport properties, initial values for plasma density, potential, and electron temperature, as well as flow parameters. As for the gas dynamics, a steady state compressible laminar flow ($Ma < 0.3$) solver is selected with no slip BC along the cathode surfaces. An inlet boundary condition (BC) with a constant mass flow rate is used, set to 0.61 mg/s of Ar gas (equivalent to 2 SCCM Xe). Ar was selected due to its simplified setup within CMS. The gas temperature is set to 1650 K, which is the selected emitter insert temperature, as will be shown next. Vacuum is represented using an outlet BC set to $p = 10$ Pa.

The CMS Plasma Module utilizes a drift-diffusion approach that requires input for ion and electron mobilities. For ions, the high electric field option is selected, incorporating charge exchange collision cross sections for Ar with Ar^+. For electrons, a reduced mobility is assumed and a Maxwellian distribution. Electron collisions with heavy particles are modeled using a set of reactions, including elastic, excitation, superelastic, ionization, Penning ionization, and

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metastable quenching processes, as detailed in Table 1. Heavy species surface reactions are also included with a sticking coefficient of 1.0 for both Ar* and Ar^+. Ion induced secondary electron emission coefficient on electrode surfaces (keeper and anode) is set to γ_i = 0.02 with emission of 2 eV mean electron energy.

Table 1. Model collisions and surface reactions.

Reaction Type	**Formula**	Collision Definition	Unit
Elastic	e+Ar => e+Ar	Cross Section	$\Delta\varepsilon = 0$ eV
Excitation	e+Ar => e+Ar*	Townsend	$\Delta\varepsilon$ = 11.5 eV
Superelastic	e+Ar* => e+Ar	Cross Section	$\Delta\varepsilon$ = -11.5 eV
Ionization	e+Ar => 2e+Ar^+	Townsend	$\Delta\varepsilon$ = 15.8 eV
Ionization	e+Ar* =>2e+Ar^+	Cross Section	$\Delta\varepsilon$ = 4.427 eV
Penning Ionization	Ar*+Ar* => e+Ar+Ar^+	Rate constant	3.734×10^8 $m^3s^{-1}mol^{-1}$
Metastable quenching	Ar*+Ar => Ar+Ar	Rate constant	1807 $m^3s^{-1}mol^{-1}$
Heavy Specie surface interaction	Ar* = >Ar	Sticking coefficient = 1.0	$\gamma_i = 0$
Heavy Specie surface interaction	Ar+=>Ar	Sticking coefficient = 1.0	$\gamma_i = 0$ or 0.02 (electron mean energy 2 eV)

Figure 1 shows schematically the simplified geometry that was used for the simulations. To reduce computational time (i.e. reducing the domain size) the keeper geometry was modified by introducing a dielectric BC representing the volume between the keeper and cathode body. The electrical BCs are set with 0 V potential on the inlet and emitter insert, and currents of I_k = 10 mA and I_a = 2 A on the keeper and anode, respectively.

To accurately resolve the steep gradients at the metal-plasma interfaces, the mesh was refined along the boundaries as well as the orifice volume. This consideration was incorporated into the mesh design with the element sizes varied across different regions of the domain: 20 µm in the orifice, 50 µm within the hollow cathode, and 100 µm in the outer domain. The computational mesh consisted of 24,000 polygonal elements in total. A sufficiently long simulation time (1 ms) was selected to ensure that all variables reached stable spatial distributions.

In the present CMS model, the thermionic emission from the emitter insert is represented by a constant thermal electron emission flux $\Gamma_t = j_{e,em}/e$, along the emitter insert surface. The value of the emission flux was determined by adjusting the flux to provide keeper potentials close to values measured experimentally, obtained for a Xe operated HHC in a keeper only configuration (i.e. anode current is zero),[6] see Table 2. The associated emitter temperature for the flux is calculated using Richardson-Dushman equation assuming a work function of 2.66 eV (for Lanthanum Hexaboride insert).[7]

Table 2. Calibration table for thermionic emission flux Γ_t as function of anode current.

I_k, A	$V_{k,exp}$, V	$V_{k,sim}$, V	Γ_t, $m^{-2}s^{-1}$	T_{em}, K
0.2	120	122	3×10^{21}	1390
0.5	55	46	3×10^{22}	1530
1.0	27	21	1×10^{23}	1620

III.Results

Two model geometries of HHC where simulated. Both with Ar gas mass flow rate of 0.61 mg/s, anode current of I_a = 2 A, floating keeper (I_k = 10 mA), and $\Gamma_t = 1.5 \times 10^{23}$ $m^{-2}s^{-1}$ (emitter at 1650 K) with a mean energy of 1 eV for emitted electrons. Each model had a different inlet diameter D_{in} whereas the keeper orifice diameter D_{ko} was kept the same, specifically: D_{in}/D_{ko} = 2 (constricted inlet) and D_{in}/D_{ko} = 4 (unconstructed inlet) ratios. Figure 2 and Fig. 3

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present two-dimensional color maps of plasma potential and electron density for each case. We observe that the plasma potential is ~35 V in the center of the orifice and emitter regions and ~ 20 V on the floating keeper electrode, a similar keeper potential was measured in past experiements.[7] Both plots show almost identical plasma potential and electron density distribution in the orifice and emitter insert regions. However, for the $D_{in}/D_{ko} = 2$ case, the plasma density and the plasma potential essentially terminate near the constriction and do not penetrate further upstream the inlet, as in the other case.

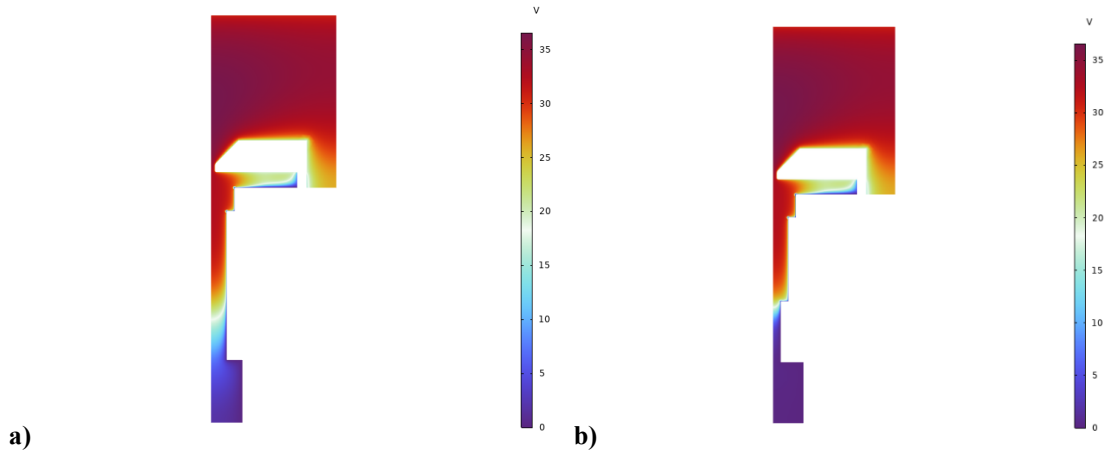

Figure 2. Calculated plasma potential distributions for $D_{in}/D_{ko} = 4$ (a) and $D_{in}/D_{ko} = 2$ (b).

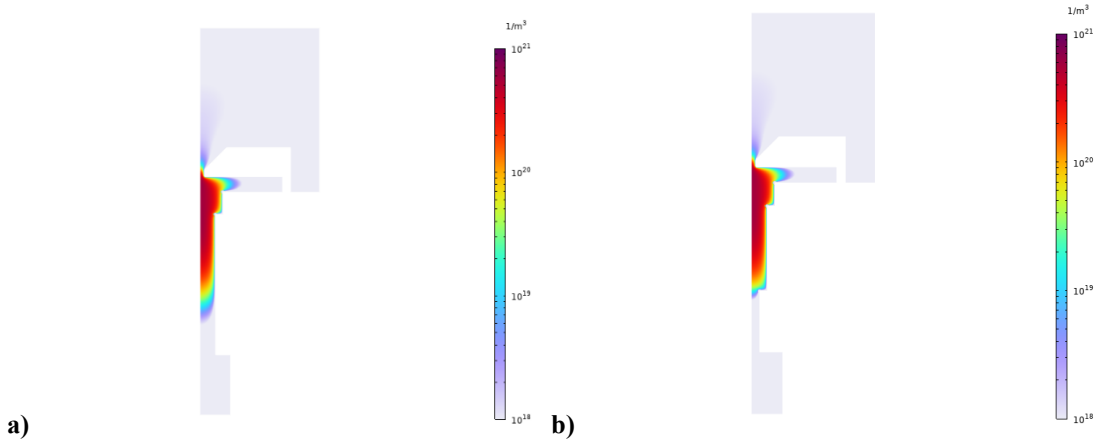

Figure 3. Calculated electron density distributions for $D_{in}/D_{ko} = 4$ (a) and $D_{in}/D_{ko} = 2$ (b).

Next, we analyze these results by comparing the spatial distribution of plasma parameters along the cathode axis. In Fig. 4a we observe that the gas pressure in the inlet section increases by 16 %. Fig. 4b shows that the gas velocity is ~ 550 m/s at the orifice. For the constricted case the flow increases speed by ~ 50 m/s in the inlet, compared to the unconstricted case.

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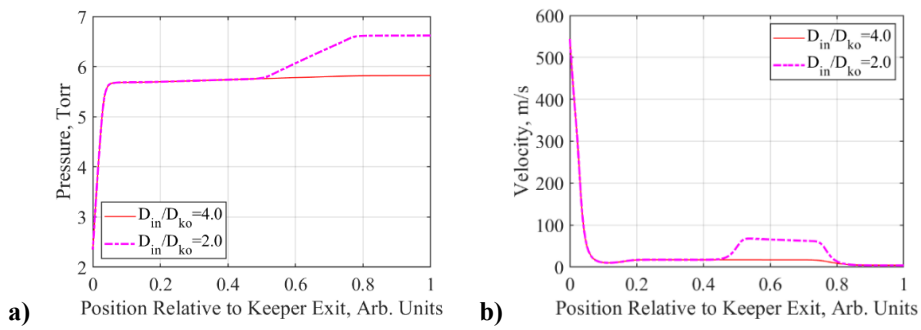

Figure 4. Calculated spatial distributions along the cathode axis for gas pressure (a) and gas velocity (b).

The gas density in the keeper orifice and emitter region is ~1.8×10^{23} m^{-3} and within the D_{in}/D_{ko} = 2 constriction it is increased by 16% (Fig. 5a). Nevertheless, we observe steep plasma density gradient upstream as shown in Fig. 5b. The electron density in the emitter insert region of ~6×10^{20} m^{-3} decays by 4 orders of magnitude between $0.5<x<0.6$, compared to only ~1.5 orders of magnitude for the unconstricted case. Figure 5c shows that the electron temperature gradually increases from the keeper orifice to the inlet $1.7<T_e<2.6$ eV. The spatial profiles are nearly identical for both cases until the constriction, at x = 0.5. Upstream x > 0.5, however, the profile diverges: in the constricted case the electron temperature reaches 4.8 eV (near the wall) and decreases farther along the inlet to below 2.6 eV; in the unconstricted case, on the other hand, presents a simple gradual lowering of T_e. The increase in electron temperature for the constricted case can be explained by need to produce sufficient ions to offset the increased wall losses.

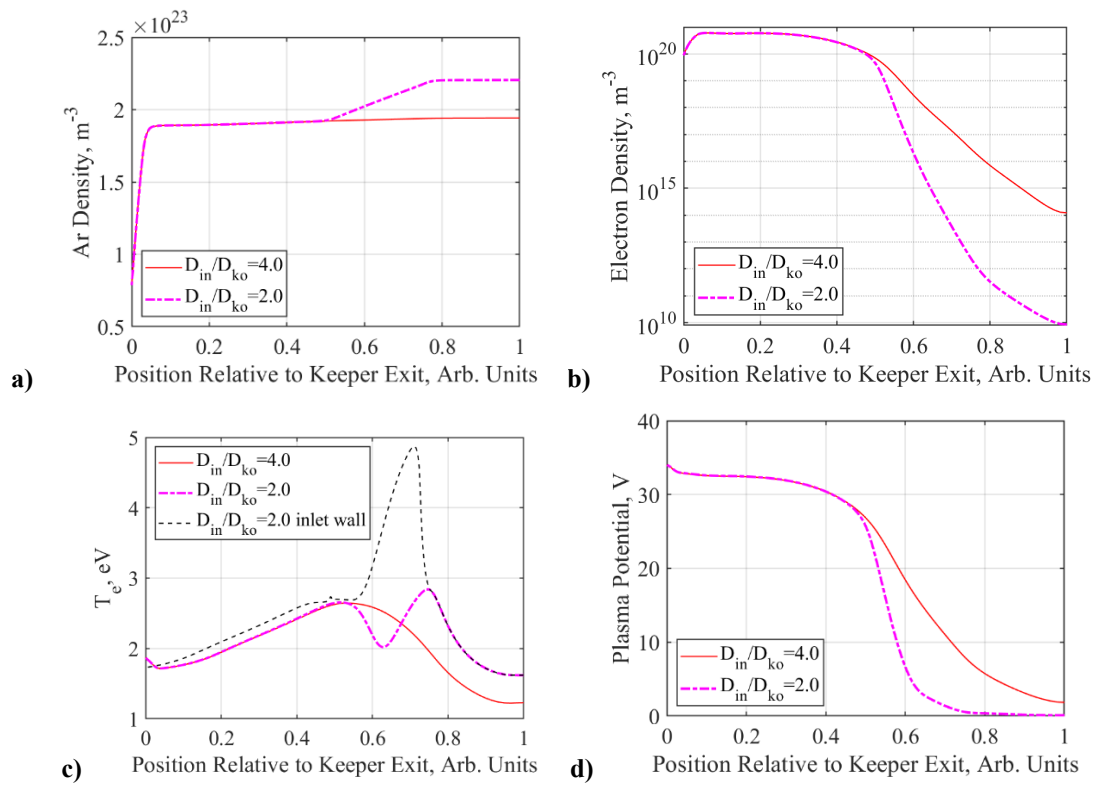

Figure 5. Spatial distribution of parameters along the axis for D_{in}/D_{ko} =2 and D_{in}/D_{ko}=4, with respect to the keeper exit (zero): gas density (a), electron density (b), electron temperature (c) plasma potential (d).

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In Fig. 5d we observe a dramatic fall in plasma potential upstream the constriction $x > 0.5$, from 35 V to about zero, whereas in the unconstricted case the potential does not reach zero at the gas input. The constriction effect on the electric field can be evaluated by Ohm's law:[8]

$$\vec{E} \backsimeq (\eta_{ei} + \eta_{en})\vec{J}_e - \nabla p_e / e n_e \backsimeq (\eta_{ei} + \eta_{en})\vec{J}_e - k_B \nabla (T_e n_e)/e n_e. \quad (1)$$

Here, η_{ei} and η_{en} are the resistivities due to electron-ion (e-i) and electron-neutral (e-n) collisions, and p_e is the electron pressure, T_e is the electron temperature and k_B is the Boltzmann constant. Resistivity is largely due to electron-ion collisions and is therefore insensitive to changes in both the plasma and neutral gas densities.[8] Therefore, the only means of regulating the electron current density $\vec{J}_e$ is through the electron pressure gradient. Thus, the large gradient in plasma density near constriction generates an increased electric field that nullifies the plasma potential in the inlet.

IV. Conclusion

We have shown, using a drift-diffusion plasma simulation, that narrowing the gas inlet diameter, near the diameter of the keeper orifice, can significantly suppress the formation of plasma in the location of the constriction. Such designs may help to eliminate unwanted Paschen breakdowns from the keeper to the gas feed.

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